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Unit 4 - Flowchart Development

PDF Documents

PDF Document 1: 1738155901510-Chapter 11 - Unit 4 - Flowchart Development.pdf

This PDF document is included in the unit materials.

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UNIT 4 FLOWCHART DEVELOPMENT

Unit Introduction

In Unit 4, you will create detailed flowcharts based on the algorithm you wrote in Unit 2 to plan and visualize the logic and steps of your smart device project. Flowcharts are essential tools that help you organize your thoughts, break down complex processes, and ensure that every part of your project works smoothly. This unit will guide you through the process of developing flowcharts for your top three ideas, allowing you to see the big picture and identify any potential issues before you start building.



DO YOU REMEMBER?

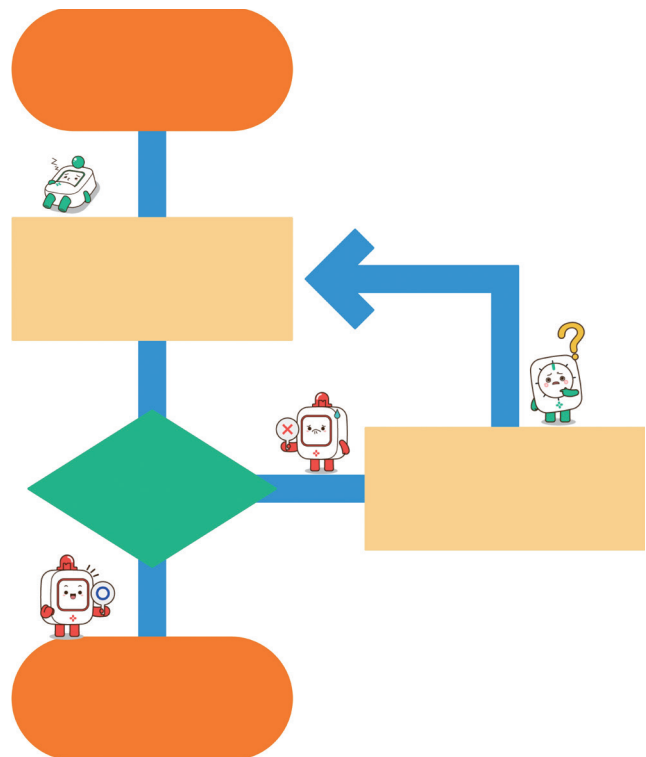
Do you remember the flowcharts you read and sometimes created in the previous chapters? Now, it's time to create your own from the ground up! Exciting!

WHY IS A FLOWCHART IMPORTANT?

Flowcharts are essential tools in planning and developing your smart device project. Here's why they are important:

With flowcharts, you can

- Organize Your Thoughts
- Visualize the Process
- Identify Issues Early
- Improve Communication
- Guide Your Work
- Enhance Problem-Solving



Developing Flowcharts

Creating Flowcharts for Your Ideas

Start by creating flowcharts for the top three ideas you selected in Unit 2. Use symbols like ovals for start and end points, rectangles for actions, diamonds for decisions, and arrows to show the flow of the process. Ensure that each step is clear and logical.



Detailing the Steps

Break down each action in your flowchart into detailed steps. Think about what needs to happen at each stage and what decisions need to be made. This will help you understand the sequence of operations and how different components interact.



Integrating MODI Modules

Incorporate the MODI modules you selected into your flowcharts. Specify which module performs each action and how they work together. This will give you a clear idea of how to program and connect the modules in your project.



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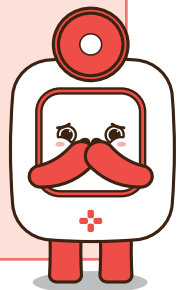


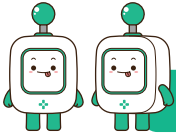
Reviewing and Refining

Review your flowcharts to check for any gaps or errors. Make sure that the logic flows smoothly and that all possible scenarios are covered. Refine your flowcharts as needed to ensure they are complete and accurate.



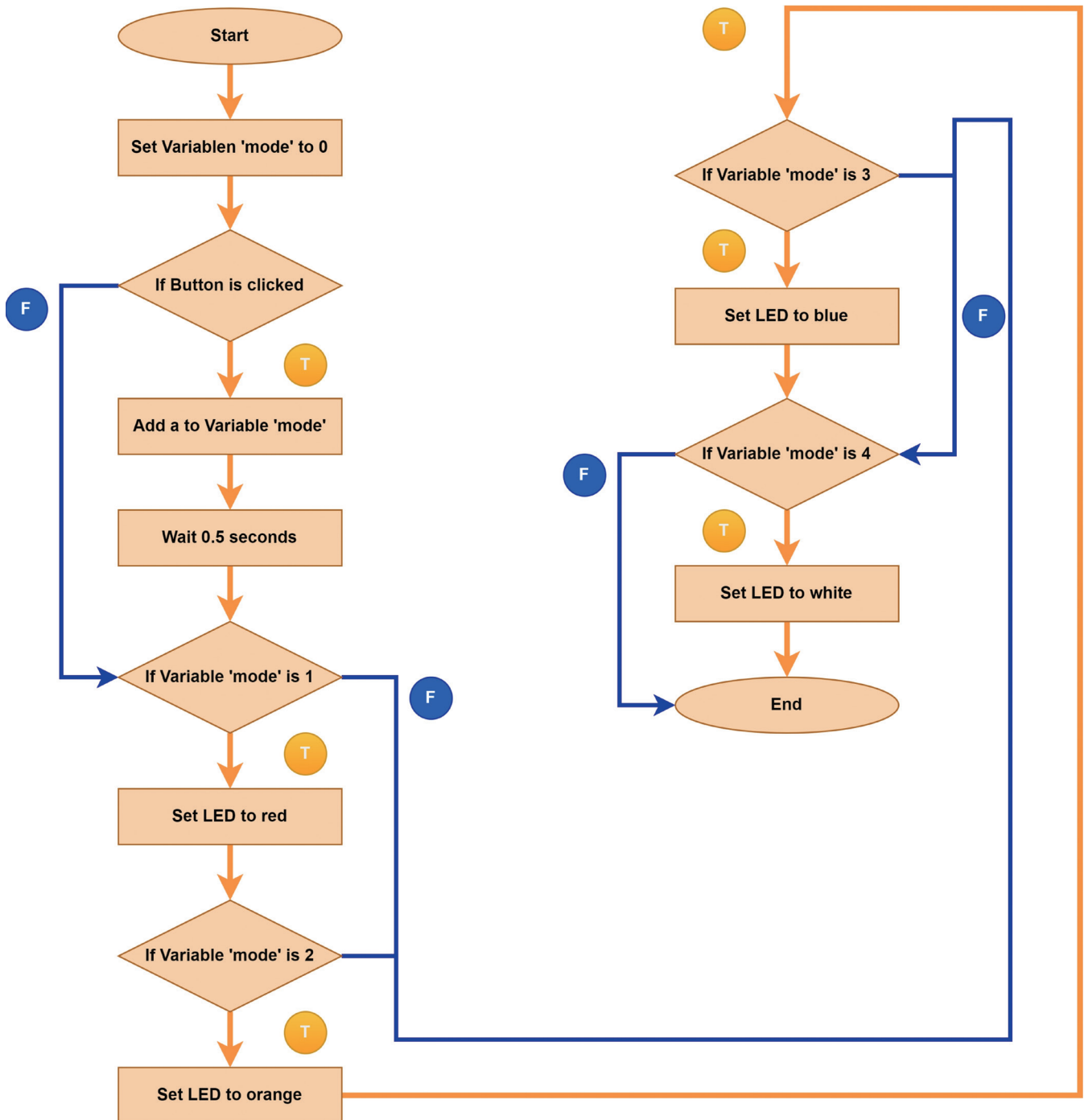
FLOWCHART BRAIN STORMING





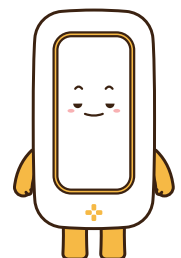
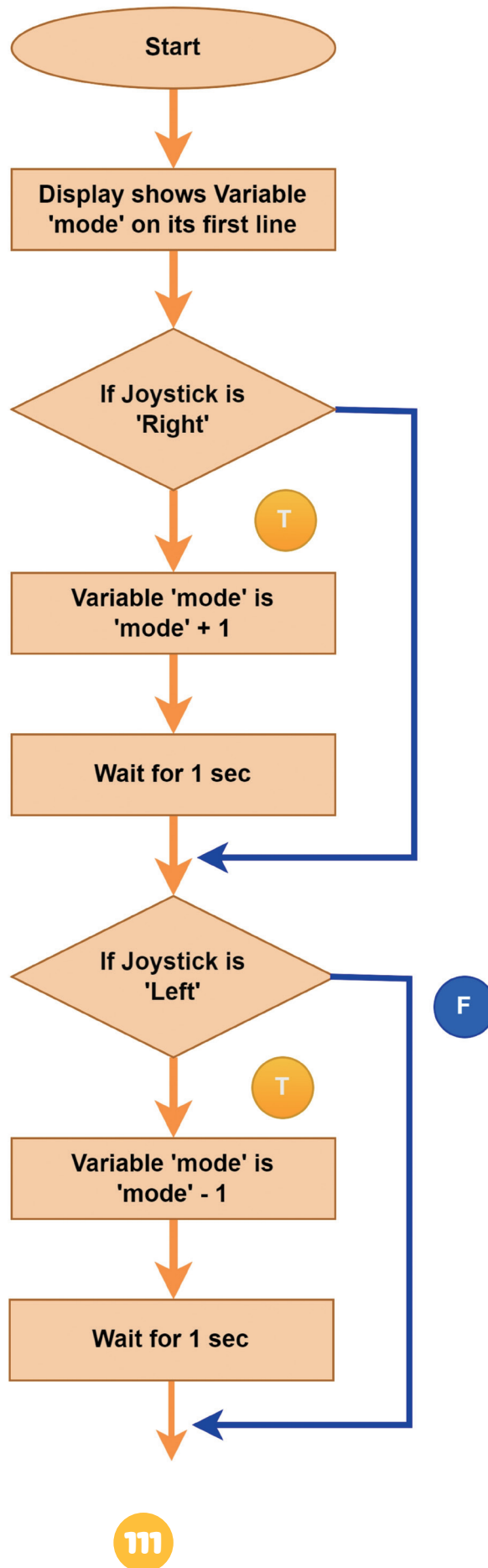
Examples of Flowchart

1





2



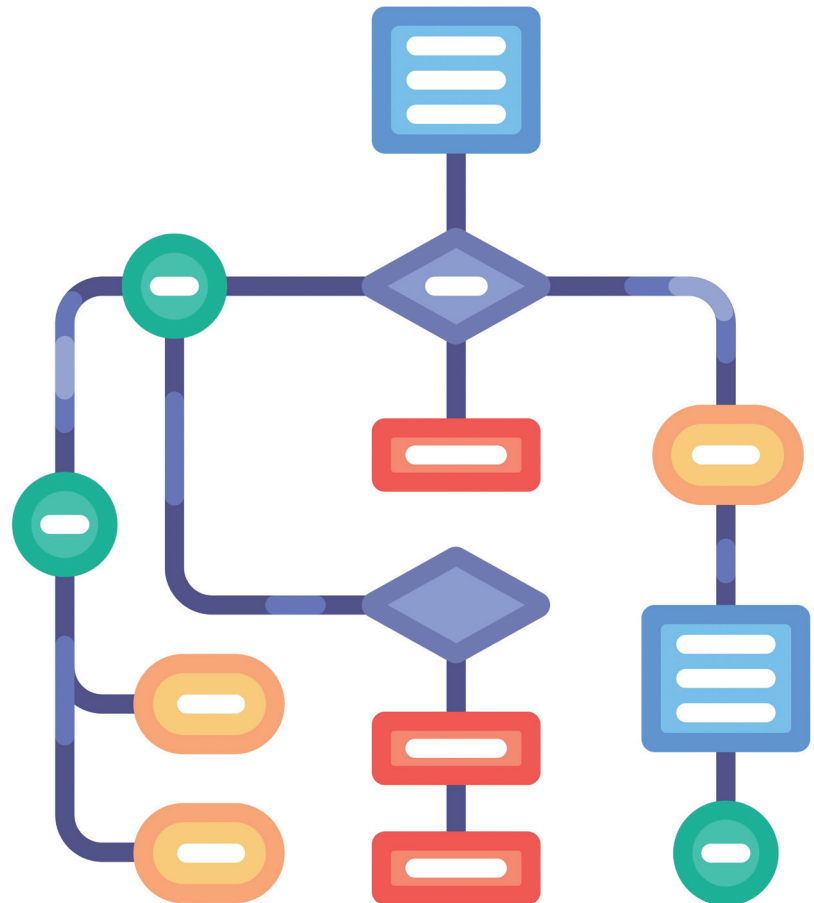
Drawing Section

Questioning the Details

- What are the key steps and decisions in your project?
- How will the MODI modules be integrated into each step of the process?
- Are there any potential issues or challenges in the flow of your project?
- What improvements can you make to ensure the flowchart is complete and accurate?



Now, it's time to draw your flowchart!





YOUR FLOWCHART



